

Battery Build-out and the Shape of Electricity Prices in California, United States

A conditional-association study of the California Independent System Operator (CAISO) SP15 day-ahead evening-minus-midday spread, May 2023–December 2025

Main finding. After adjustment for matched-window demand and solar patterns, delivered natural-gas cost, calendar effects and a linear trend, the battery coefficient is $\hat{\beta} = 0.645$ USD/MWh per GW with $SE_{HAC(7)} = 8.012$ and $p = 0.936$. The estimate is too imprecise to establish either a narrowing or a widening of the spread.

1. Research question and scope

California's battery fleet expanded rapidly while the state's solar-rich midday and evening ramp produced a strong intertemporal price signal. The research question is whether greater operating utility-scale battery power capacity was associated with a narrower daily evening-minus-midday day-ahead electricity-price spread at the CAISO SP15 generation trading hub, after controlling for the daily demand pattern, solar pattern, Southern California delivered natural-gas costs, calendar seasonality and a time trend. The estimand is a conditional historical association: the expected difference in the spread associated with 1 GW more capacity, holding the included regressors fixed. It is not a causal policy effect, a battery-profit calculation or an optimisation of dispatch.

2. Hypotheses

The conventional evidence standard is a two-sided test because either sign is economically possible once deployment responses and market evolution are considered.

$$H_0: \beta = 0 \quad \text{versus} \quad H_1: \beta \neq 0$$

The pre-specified economic prediction is $\beta < 0$: more usable storage should compress the evening-minus-midday spread. The sign prediction is reported separately from the two-sided test to avoid selecting a one-sided test after observing the estimate. For the mechanism regressions, charging and discharging imply $\beta_E < 0$ for the evening-price component and $\beta_M > 0$ for the midday-price component, so $\beta_{\text{spread}} = \beta_E - \beta_M < 0$. A non-significant or positive estimate remains a valid result and must not be rewritten as evidence of no economic role for batteries.

3. Intuition behind the econometric baseline model

Solar generation tends to reduce net load and prices around midday. In the evening, solar output falls while demand can remain high. A battery can charge in the low-price period, adding demand and tending to raise the midday price, then discharge later, adding supply and tending to lower the evening price. Both movements should narrow the spread. This is the storage-arbitrage mechanism discussed by Sioshansi et al. (2009) and observed in CAISO battery operating patterns. The measured spread is an intertemporal price signal, not realised profit: it omits efficiency losses, degradation, operating costs, quantities dispatched and ancillary-service revenue.

For Pacific-time day t , the pre-specified hourly windows and daily outcome are:

$$\begin{aligned} H_E &= \{17,18,19,20\}, & H_M &= \{10,11,12,13,14,15\} \\ \bar{P}_t^E &= \frac{1}{4} \sum_{h \in H_E} P_{th}, & \bar{P}_t^M &= \frac{1}{6} \sum_{h \in H_M} P_{th} \\ Y_t &= \text{Spread}_t = \bar{P}_t^E - \bar{P}_t^M \end{aligned}$$

Matched-window demand and solar controls separate the battery association from mechanical changes in the evening ramp. The gas-cost proxy controls a major marginal-generation input. Weekday and month indicators absorb recurring calendar patterns, while elapsed time absorbs a smooth common trend. The baseline is therefore:

$$\begin{aligned} Y_t &= \alpha + \beta B_t + \gamma_1 \text{DemandGap}_t + \gamma_2 \text{SolarDrop}_t \\ &+ \gamma_3 G_t + \sum_{d=2}^7 \delta_d D_{dt} + \sum_{m=2}^{12} \mu_m M_{mt} + \lambda T_t + u_t \end{aligned}$$

Here B_t is month-start battery power capacity in GW and β is the partial slope in USD/MWh per GW. Monday and January are the omitted calendar categories. The spread remains in levels because it can be zero or negative, so a logarithm is not defined for every observation.

4. Data: obtaining, cleaning and processing

All datasets are obtainable: SP15 day-ahead total LMPs come from CAISO OASIS; utility-scale battery inventories come from EIA-860M; adjusted hourly demand and solar generation come from EIA-930; and the FRSCE3 Southern California delivered gas-cost proxy comes from CAISO's PRC_FUEL data. The archived package contains the compact source extracts, merged daily file, preparation script, analysis script, data dictionary, machine-readable outputs and reproducibility audit. A second execution of the two-script workflow reproduced identical hashes for all 20 generated CSV outputs.

Table 1. Data sources and analytical construction

Variable	Construction	Official source
SP15 spread	Average 17:00–20:59 total LMP minus average 10:00–15:59 total LMP; USD/MWh	CAISO OASIS
Battery power	Operating CAISO battery nameplate MW; prior month-end assigned to next month; divided by 1,000	EIA-860M
Demand and solar	Evening-minus-midday demand gap and midday-minus-evening solar drop; GW	EIA-930
Gas cost	Daily non-GHG Southern California delivered-cost proxy; USD/MMBtu	CAISO PRC_FUEL, FRSCE3

Notes: CAISO-wide demand and solar are imperfect geographic matches for SP15. FRSCE3 includes transportation and in-kind reimbursement and is not the stand-alone ICE commodity index.

The planned window contains 976 Pacific-time dates from 1 May 2023 through 31 December 2025. Source timestamps were preserved, aligned to the interval-start convention and converted once to America/Los_Angeles before aggregation. Unique date keys and duplicates were audited separately for every source. EIA hour-ending observations 11–16 were aligned to price starts 10–15, and hour endings 18–21 to price starts 17–20. Capacity was held constant within each month using the previous month-end inventory, preventing look-ahead.

A day enters the regression only when all four evening and six midday price, demand and solar observations are present. Official adjusted EIA-930 demand and solar fields are blank for all 24 hours on 2 November 2024; no value was imputed, leaving 975 complete days. Adjusted solar categories from July 2024 onward were combined consistently. Total LMP—not only its energy, congestion or loss component—was retained. Genuine negative prices and scarcity spikes were not winsorised; ten days with absolute externally studentised residuals above three were flagged for source review and analysed only in a labelled sensitivity.

5. Assumptions and econometric method

Conditional mean. A slope interpretation requires the remaining disturbance to satisfy the stated orthogonality condition after conditioning on the included regressors.

$$E(u_t | X_t) = 0$$

Timing and measurement. Capacity must represent availability before the daily price outcome, timestamps must refer to the same operating day, and remaining measurement error must not create systematic correlation with the disturbance.

Functional form and rank. The linear model is treated as a best linear projection. No exact collinearity is expected, but near collinearity can make estimates extremely imprecise. Stable coefficients over the sample are assumed for this descriptive specification.

Inference. Daily errors may be heteroskedastic and autocorrelated. OLS coefficients are retained, but the reporting standard is Newey–West HAC covariance with seven daily lags. HAC changes estimated uncertainty; it does not remove omitted dynamics or create exogenous deployment.

Three nested models show how conditioning changes the battery coefficient: capacity only; capacity plus demand gap, solar drop and gas cost; and the full baseline adding weekday effects, month effects and trend. Diagnostics include residual-versus-fitted and residual-over-time plots, a 30-day residual autocorrelation function, Durbin–Watson, a seven-lag Breusch–Godfrey test, correlations, VIFs, augmented Dickey–Fuller tests and influence diagnostics. Focused checks use an additional capacity lag, energy rather than power capacity, component-price outcomes, the FRSCE3 tariff break, monthly first differences, an outlier exclusion and negative-price hours. These are robustness checks, not a search for significance.

6. Findings

The mean spread is 45.18 USD/MWh, the median is 38.56 and the standard deviation is 41.48; the range is –0.11 to 841.08. Operating battery power capacity rises from 4.856 GW to 15.435 GW. In the capacity-only model, 1 GW more capacity is associated with a 3.755 USD/MWh narrower spread (HAC SE 0.805, $p < 0.001$). Adding fundamentals gives –2.765 (SE 0.598, $p < 0.001$). After calendar effects and trend are added, the coefficient changes to 0.645 (SE 8.012, $p = 0.936$), with a 95% confidence interval from –15.059 to 16.349. The baseline R-squared is 0.286. The sign change indicates sensitivity to common time movement, not evidence that batteries widen spreads.

Table 2. Full baseline coefficients with seven-lag Newey–West standard errors

Regressor	Coefficient	HAC SE	p-value	95% confidence interval
Battery power capacity (GW)	0.645	8.012	0.936	[–15.059, 16.349]
Demand gap (GW)	4.449	1.244	<0.001	[2.012, 6.887]
Solar drop (GW)	1.571	0.621	0.012	[0.353, 2.788]
FRSCE3 gas cost	9.534	2.639	<0.001	[4.361, 14.707]
Linear trend (years)	–11.539	33.068	0.727	[–76.351, 53.273]

Notes: Dependent variable is the SP15 evening-minus-midday day-ahead spread in USD/MWh. Weekday and month indicators are included but omitted from the compact display. N = 975.

The fundamentals have the expected conditional signs: larger evening demand, a larger solar drop and higher delivered gas cost are associated with wider spreads. They are controls, not separately identified causal effects. The battery estimate is economically small at its centre but the uncertainty interval is economically wide. A non-significant result therefore means weak precision, not a tightly estimated zero.

The central precision problem is the 0.9975 correlation between battery capacity and trend; their VIFs are 391.3 and 387.2. Although there are 975 daily outcomes, capacity changes mainly across only 32 monthly levels. Serial dependence is also material: the Durbin–Watson statistic is 0.918, and the seven-lag Breusch–Godfrey F statistic is 63.42 with $p < 0.001$. The residual autocorrelation is about 0.54 at one day. These results justify HAC inference but also show that the nominal daily sample size overstates independent information. ADF checks with constant, trend and seven lagged differences reject a unit root for each principal series, but this does not solve identification.

Table 3. Focused checks of the battery association

Specification or outcome	Estimate (HAC SE)	p-value	N
Full baseline [GW]	0.645 (8.012)	0.936	975
Additional one-month lag [GW]	2.271 (6.524)	0.728	944
Exclude studentised residual > 3 [GW]	-0.787 (4.952)	0.874	965
Add FRSCE3 tariff-break indicator [GW]	4.791 (8.226)	0.560	975
Monthly first difference [change in GW]	-21.017 (17.094)	0.219	31
Energy-capacity alternative [GWh]	1.588 (0.844)	0.060	975
Evening LMP outcome [GW]	4.010 (8.985)	0.655	975
Midday LMP outcome [GW]	3.366 (5.256)	0.522	975
Negative-price hours outcome [GW]	-1.403 (0.797)	0.078	975

Notes: Brackets identify the battery unit. The monthly first-difference estimate uses 31 monthly changes. Confidence intervals for all displayed checks include zero.

No focused check overturns the baseline conclusion. The extra timing lag remains imprecise. Removing ten genuine extreme-residual days changes the sign but not the inference. The energy-capacity and negative-price-hour estimates are marginal at the 10% level but their 95% intervals include zero and they face the same common-trend problem. Component regressions do not show the predicted evening decrease and midday increase precisely.

7. Research limitations

First, deployment is not random. Expected arbitrage opportunities can attract investment, producing reverse association. Solar additions, thermal retirements, transmission upgrades, hydrology, imports, outages, carbon costs, market rules and policy may move with both capacity and prices. Calendar effects and a trend cannot guarantee that these omitted influences are removed. Second, the near-perfect capacity–trend relationship leaves little independent variation for separating build-out from time. Dropping the trend merely to improve significance would change the identifying assumption and risk attributing gradual market evolution to batteries.

Third, EIA-860M measures preliminary nameplate utility-scale capacity, not actual battery operation. It omits sub-1-MW and behind-the-meter systems and does not reveal state of charge, duration, degradation, outages, bidding, dispatch or ancillary-service participation. Fourth, CAISO-wide demand and solar are imperfect geographic matches for the southern SP15 hub, while FRSCE3 is a delivered-cost proxy rather than the stand-alone commodity index. Fifth, extreme prices create heteroskedasticity and influential observations, and residual dependence indicates omitted short-run dynamics. Finally, the May 2023–December 2025 window is short and contains only 32 monthly exposure levels. A credible causal study would require longer independent variation, plausibly exogenous deployment or operating shocks, or direct dispatch data—not simply more controls.

8. Practical implications

The evidence does not support a claim that each additional GW of installed battery capacity reliably narrowed the SP15 spread during this period. It also does not show that batteries had no influence: the

confidence interval includes economically meaningful narrowing and widening. Policymakers and market participants should therefore avoid using the point estimate as a forecast of a storage mandate. Installed MW should be supplemented with state-of-charge, availability, duration, bidding and dispatch data before drawing operational conclusions.

For research design, the strongest practical lesson is to seek variation in actual operation or plausibly exogenous deployment timing rather than relying on a monotonic capacity trend. For market monitoring, the pronounced scarcity episodes, negative midday prices and residual persistence warrant separate operational analysis. A narrower energy-price spread is only one benefit channel: reliability, emissions, congestion, consumer costs and ancillary services are outside this outcome. The present result should therefore guide better measurement and identification, not be treated as a verdict on storage value.

9. Conclusion

The fully adjusted association is 0.645 USD/MWh per GW with 95% CI = $[-15.059, 16.349]$. The data cannot distinguish the independent battery association from zero or from contemporaneous market change. Severe capacity–trend collinearity, serial dependence and non-random deployment are the decisive limits; the appropriate conclusion is conditional and historical, not causal.

References

California ISO. n.d. OASIS: Open Access Same-time Information System. [CAISO OASIS portal](#)

California ISO Department of Market Monitoring. 2025. 2024 Annual Report on Market Issues and Performance. [CAISO DMM report](#)

Sioshansi, R., Denholm, P., Jenkin, T. and Weiss, J. 2009. 'Estimating the value of electricity storage in PJM: Arbitrage and some welfare effects.' *Energy Economics* 31(2): 269–277. [DOI record](#)

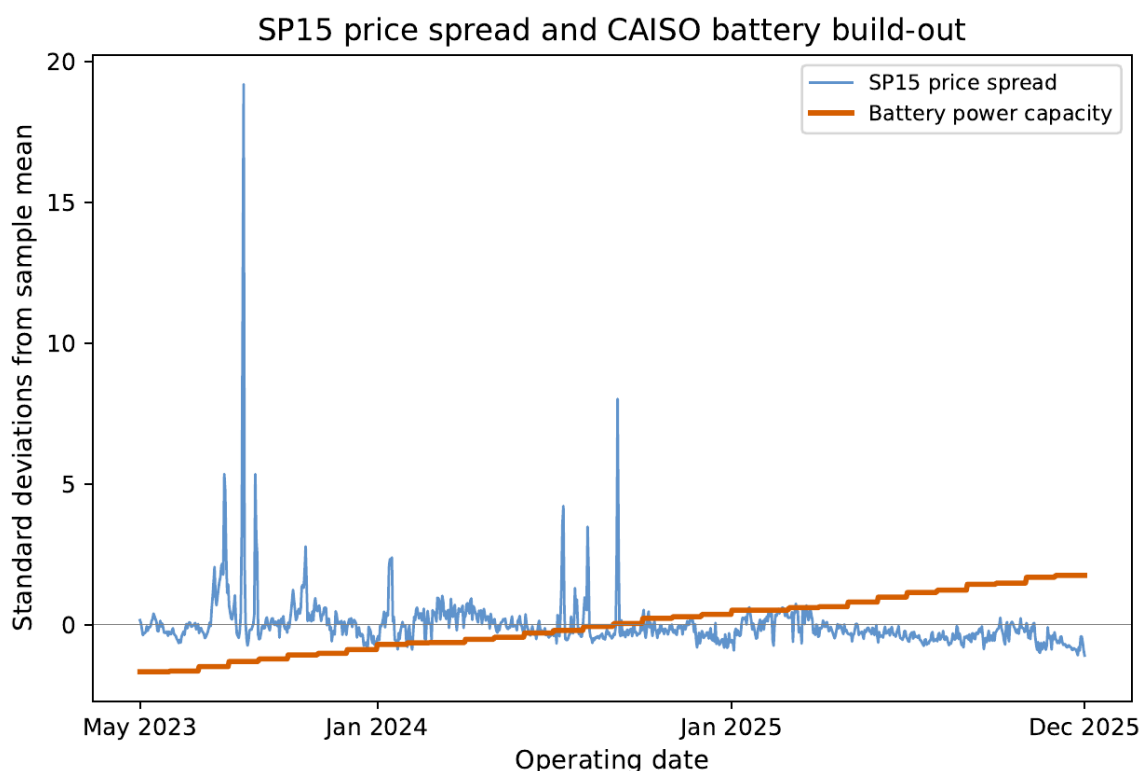
U.S. Energy Information Administration. n.d. EIA-860M Preliminary Monthly Electric Generator Inventory. [EIA-860M data page](#)

U.S. Energy Information Administration. n.d. EIA-930 Hourly Electric Grid Monitor. [EIA-930 information page](#)

Appendix. Complete figure set and interpretive comments

The seven figures below are reproduced from the validated final-analysis package. Each comment explains what the visual contributes and what it cannot establish.

Appendix Figure A1. Daily SP15 spread and operating battery power capacity, standardised for visual comparison



Comment. Battery capacity increases almost monotonically, from 4.856 GW to 15.435 GW, while the standardised spread has large short-lived spikes and becomes generally lower later in the sample. The visual supports the simple negative association, but it also exposes the main identification problem: capacity is nearly a time index. Any gradual change in solar penetration, generation mix, transmission, fuel conditions or market rules can resemble a battery relationship. The isolated spread spikes show why daily averages and standardisation must not be mistaken for stable treatment variation. The regression uses the variables in their original units and introduces calendar effects, fundamentals and trend precisely because this visual correlation is not causal evidence.

Source: validated final-analysis package, figure_01_capacity_and_price.pdf. Regressions and comments use the real 975-day analysis sample.

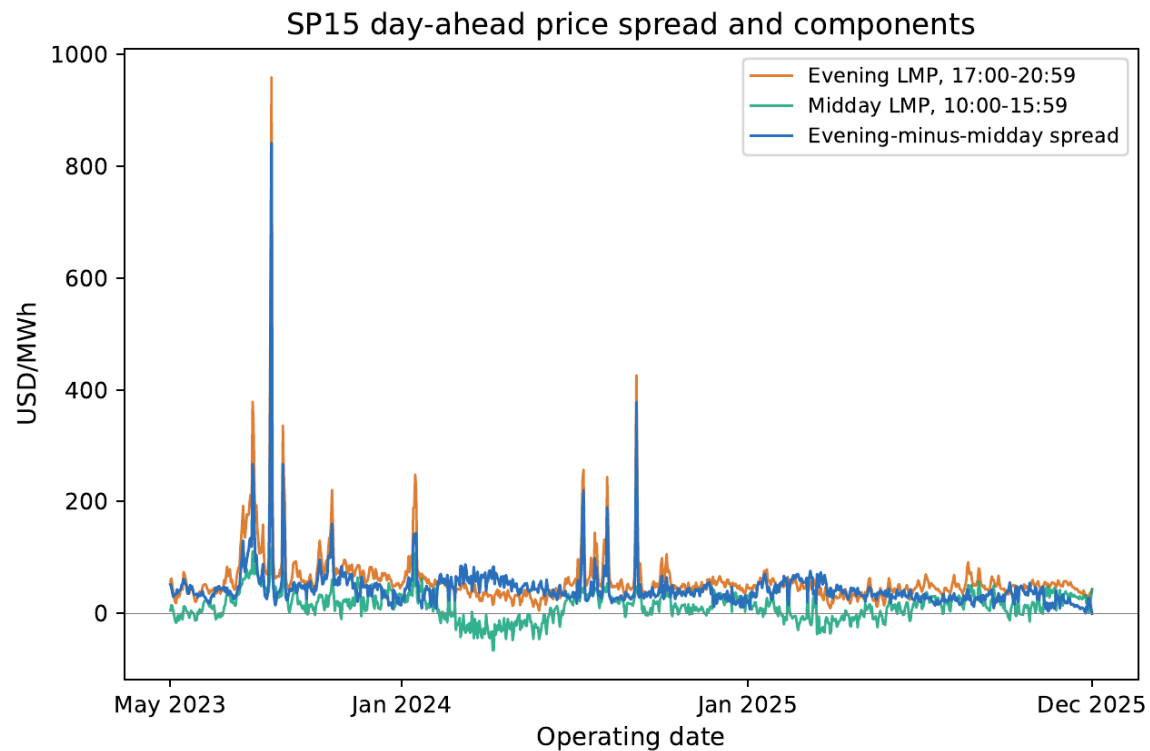
Data sources:

EIA-860M: <https://www.eia.gov/electricity/data/eia860m/>

EIA-930: <https://www.eia.gov/electricity/gridmonitor/about>

CAISO Monthly Renewables Performance Reports: <https://www.caiso.com/library/monthly-renewables-performance-report>

Appendix Figure A2. SP15 evening LMP, midday LMP and their daily spread



Comment. The evening price generally exceeds the midday price, producing a positive spread on most days. Large spread episodes are driven mainly by extreme evening prices, especially in August 2023 and September 2024, while midday prices sometimes fall below zero. This pattern is consistent with the economic mechanism—solar-rich midday conditions and tighter evening conditions—but it does not isolate batteries. It also shows why the outcome is kept in levels and why genuine extremes are retained: a logarithm is not defined for non-positive values, and scarcity events are economically meaningful rather than automatic data errors. The component regressions remain imprecise, so the figure illustrates the market pattern without confirming the predicted battery channel.

Source: validated final-analysis package, figure_02_price_components.pdf. Regressions and comments use the real 975-day analysis sample.

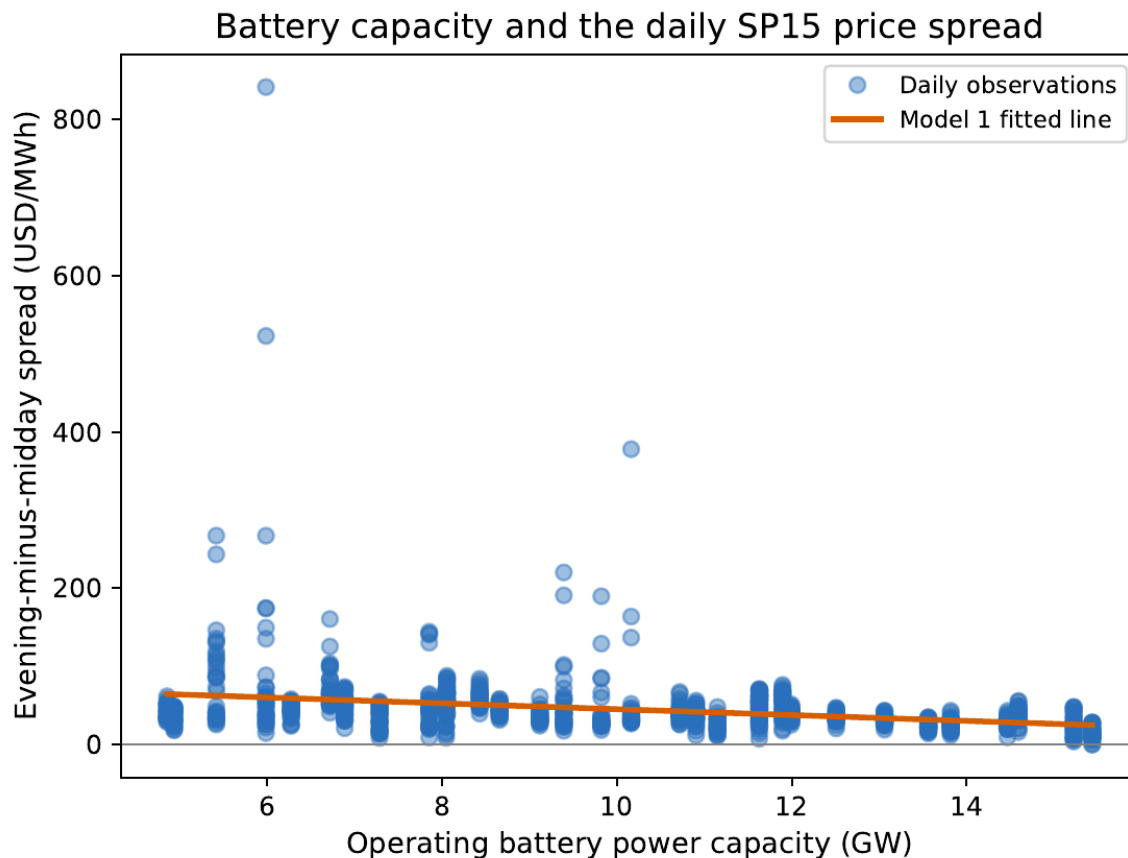
Data sources:

EIA-860M: <https://www.eia.gov/electricity/data/eia860m/>

EIA-930: <https://www.eia.gov/electricity/gridmonitor/about>

CAISO Monthly Renewables Performance Reports: <https://www.caiso.com/library/monthly-renewables-performance-report>

Appendix Figure A3. Battery power capacity and the daily SP15 spread



Comment. The downward capacity-only fitted line corresponds to a coefficient of -3.755 USD/MWh per GW, with a seven-lag HAC standard error of 0.805. However, the vertical bands reveal that the apparent 975-observation scatter contains only 32 monthly capacity levels. Several high-price days dominate the vertical scale, and the fitted line combines battery build-out with every other development that moves over time. After demand, solar, gas cost, calendar effects and trend are included, the coefficient changes to 0.645 and becomes extremely imprecise. The figure is therefore useful as a descriptive starting point and as a warning: the raw negative slope should not be interpreted as the independent or causal effect of batteries.

Source: validated final-analysis package, figure_03_battery_spread_scatter.pdf. Regressions and comments use the real 975-day analysis sample.

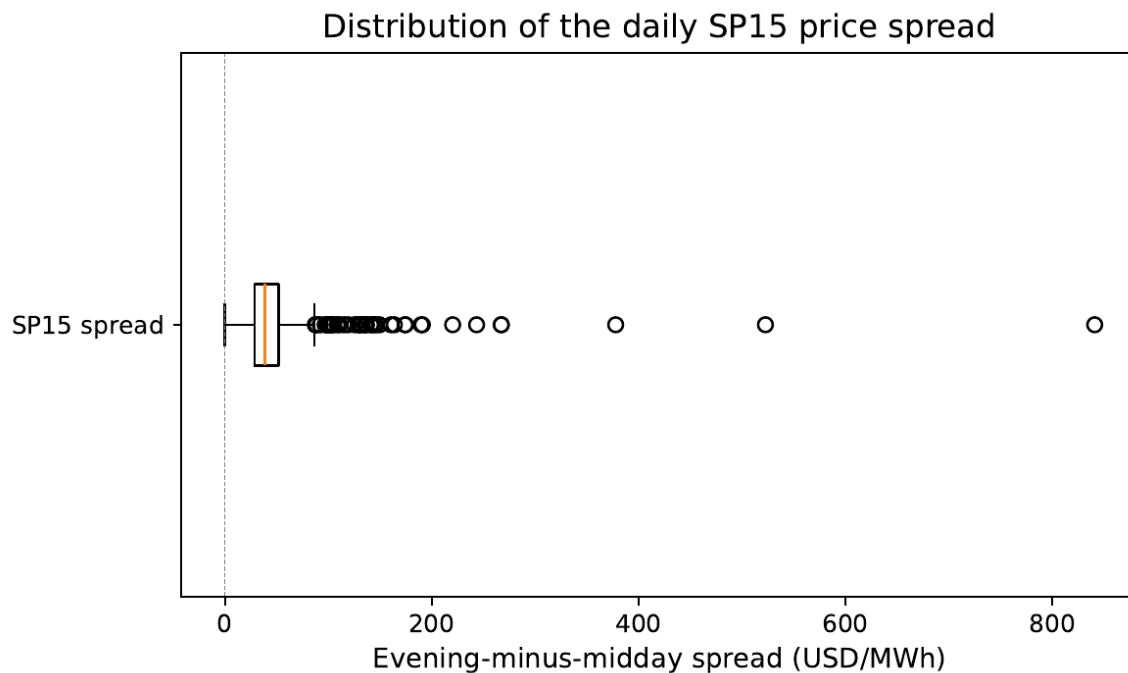
Data sources:

EIA-860M: <https://www.eia.gov/electricity/data/eia860m/>

EIA-930: <https://www.eia.gov/electricity/gridmonitor/about>

CAISO Monthly Renewables Performance Reports: <https://www.caiso.com/library/monthly-renewables-performance-report>

Appendix Figure A4. Distribution of the daily SP15 evening-minus-midday spread



Comment. The median spread is 38.56 USD/MWh and the mean is 45.18, while the maximum reaches 841.08 and the minimum is -0.11. The mean above the median and the long sequence of upper-tail points show pronounced right skew. These observations are verified scarcity-price episodes, not transcription errors, so the primary outcome is not winsorised. Their presence can create heteroskedasticity, affect least-squares precision and make normal-looking central summaries incomplete. The labelled sensitivity excluding ten days with absolute externally studentised residuals above three changes the battery estimate to -0.787 but leaves it non-significant. Thus, the main conclusion is not solely an artefact of the most extreme days, although those days materially shape the distribution.

Source: validated final-analysis package, figure_04_spread_boxplot.pdf. Regressions and comments use the real 975-day analysis sample.

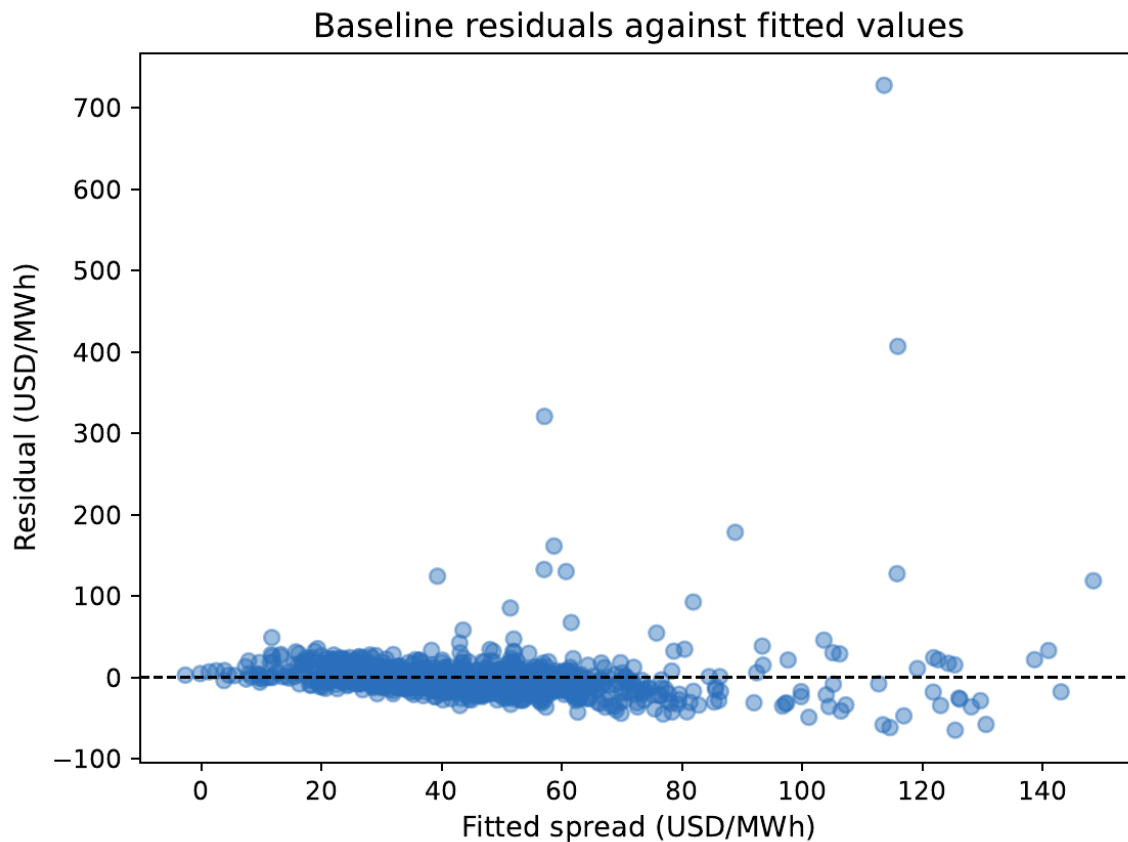
Data sources:

EIA-860M: <https://www.eia.gov/electricity/data/eia860m/>

EIA-930: <https://www.eia.gov/electricity/gridmonitor/about>

CAISO Monthly Renewables Performance Reports: <https://www.caiso.com/library/monthly-renewables-performance-report>

Appendix Figure A5. Baseline residuals against fitted values



Comment. The residual cloud is clearly non-uniform. Most fitted spreads lie below approximately 80 USD/MWh, where residuals are comparatively concentrated around zero, but their dispersion changes with the fitted value and several very large positive residuals remain. This pattern is consistent with heteroskedasticity and indicates that the baseline model underpredicts some high-spread or scarcity episodes. Together with the documented serial dependence, it supports the pre-specified use of seven-lag Newey–West covariance rather than conventional OLS standard errors. HAC inference, however, adjusts only the estimated covariance matrix; it does not correct misspecification of the conditional mean, remove influential observations, or resolve omitted-variable and regime-specific effects. The plot also raises, but does not establish the possibility of functional-form misspecification. A useful additional diagnostic would be to superimpose a LOESS or comparable nonparametric smooth. A smooth remaining close to zero would be more consistent with changing residual variance than with systematic mean misspecification; persistent curvature or departure from zero would strengthen the case for exploring nonlinearities, interactions or regime-dependent covariates. Because the observations are serially dependent, any confidence bands should account for that dependence, and any revised specification should be reported as an exploratory robustness check rather than selected for statistical significance. Verified scarcity observations should not be deleted automatically, because they may contain economically important information that the baseline model does not capture.

Source: validated final-analysis package, figure_05_residuals_fitted.pdf. Regressions and comments use the real 975-day analysis sample.

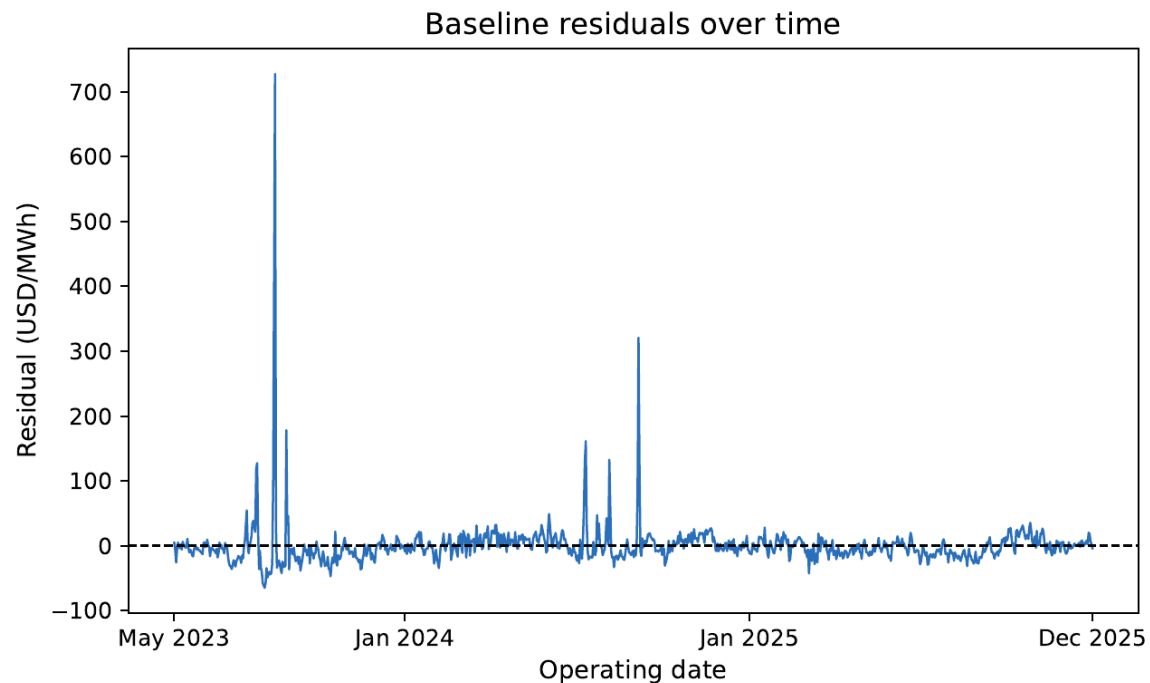
Data sources:

EIA-860M: <https://www.eia.gov/electricity/data/eia860m/>

EIA-930: <https://www.eia.gov/electricity/gridmonitor/about>

CAISO Monthly Renewables Performance Reports: <https://www.caiso.com/library/monthly-renewables-performance-report>

Appendix Figure A6. Baseline residuals over time



Comment. Residuals occur in runs rather than as featureless day-to-day noise. The largest positive residual appears in August 2023, with further clusters in mid-2024 and September 2024. These are dates on which the included fundamentals, calendar effects and linear trend underpredict the realised spread. Source checks found no reason to classify the flagged observations as errors. Their temporal clustering is consistent with unmodelled short-run market conditions such as outages, heat, imports, hydro availability or network constraints. The time pattern motivates serial-correlation tests and HAC inference and reinforces the study's associational scope. It also suggests that richer operational controls or a carefully specified dynamic model could improve future work, provided such extensions are pre-specified and do not obscure identification.

Source: validated final-analysis package, figure_06_residuals_time.pdf. Regressions and comments use the real 975-day analysis sample.

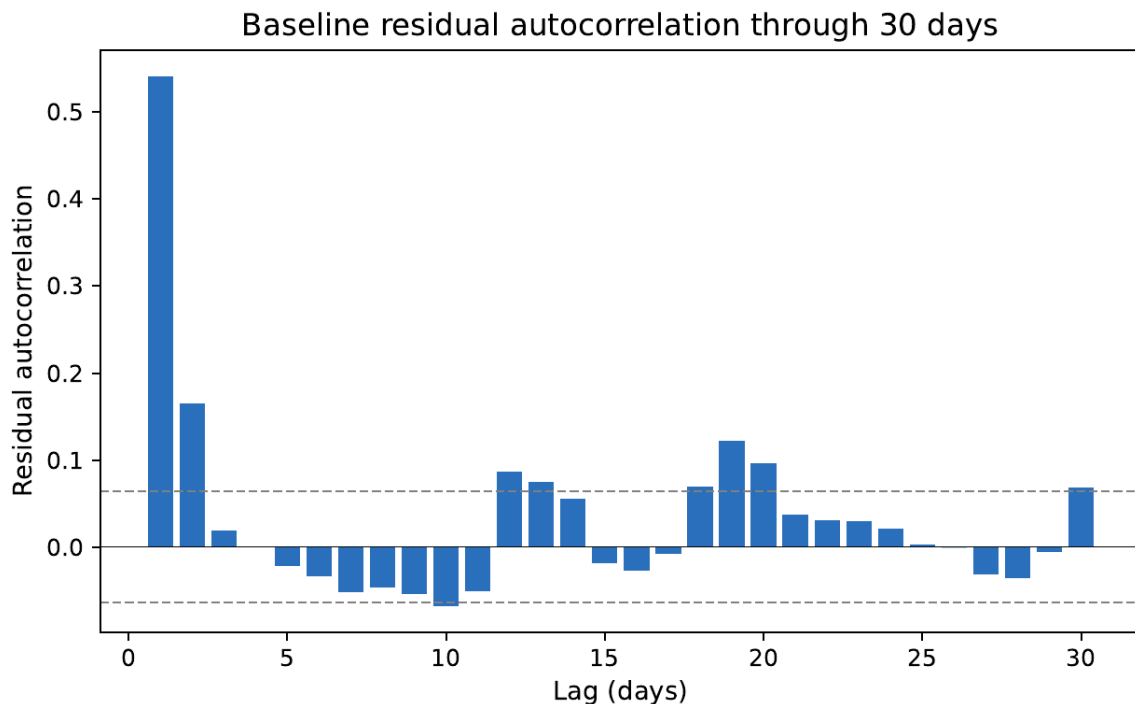
Data sources:

EIA-860M: <https://www.eia.gov/electricity/data/eia860m/>

EIA-930: <https://www.eia.gov/electricity/gridmonitor/about>

CAISO Monthly Renewables Performance Reports: <https://www.caiso.com/library/monthly-renewables-performance-report>

Appendix Figure A7. Baseline residual autocorrelation through 30 days



Comment. The first-order residual autocorrelation is approximately 0.54, and multiple later lags lie outside the approximate 95% reference bands. This agrees with the Durbin–Watson statistic of 0.918 and the seven-lag Breusch–Godfrey F statistic of 63.42 with $p < 0.001$. Conventional independent-error standard errors are therefore unsuitable. The report uses seven-lag Newey–West inference as pre-specified, while recognising that the ACF contains dependence beyond one week. The visual does not invalidate OLS coefficients mechanically, but it reduces confidence in naive precision and indicates omitted temporal structure. HAC inference is a correction to uncertainty, not a cure for omitted dynamics, common trends or non-random storage deployment.

Source: validated final-analysis package, figure_07_residual_acf.pdf. Regressions and comments use the real 975-day analysis sample.

Data sources:

EIA-860M: <https://www.eia.gov/electricity/data/eia860m/>

EIA-930: <https://www.eia.gov/electricity/gridmonitor/about>

CAISO Monthly Renewables Performance Reports: <https://www.caiso.com/library/monthly-renewables-performance-report>